CSE 4345: Big Data Analytics

Week 9, Part 2 — Seminal Research & Case Study

Department of Computer Science & Engineering
Premier University, Chittagong

Semester 2026

Part 2 Overview — Week 9

Today's Agenda

Seminal Works Covered

1. Tufte, E. R. (1983). *The Visual Display of Quantitative Information.* Graphics Press.
2. Shneiderman, B. (1996). *The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations.* IEEE Visual Languages.

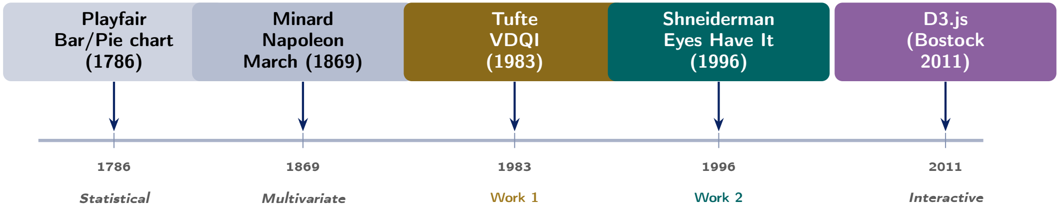
Case Study

New York Times Graphics Desk (2010–2022): interactive data journalism using D3.js, reaching **6 million readers per day**; COVID-19 visualisations cited in policy briefs worldwide.

CLO Addressed

Work 1 — Tufte (1983): The Visual Display of Quantitative Information

Research Landscape: History of Data Visualisation



Why Tufte Matters for Big Data Visualisation

Big Data pipelines produce outputs humans must interpret. A chart with **chartjunk** on a 4K dashboard misleads as badly as a bad statistical test. Tufte's principles are the analytical foundation for every modern dashboard framework (Tableau, Grafana, D3). The week's D3 practicals are applications of Tufte's rules.

Tufte (1983): Publication Context

Full Citation

Tufte, E. R. (1983). **The Visual Display of Quantitative Information.** Graphics Press, Cheshire, CT. (2nd ed., 2001; 197 pp.)

Format: Book (self-published by Tufte; designed and typeset by Tufte himself)
Cited by: >20,000 (Google Scholar, 2024)
Author: Edward R. Tufte, Professor of Statistics, Graphic Design, and Political Economy (Yale University)

Note: the book was *rejected by every publisher* Tufte approached; he mortgaged his house to self-publish. It sold 40,000 copies in its first year.

Tufte's Central Claim

The Context: Why 1983?

- By 1983, computer-generated charts were proliferating in corporate reports and newspapers
- Most were produced by software that defaulted to heavy gridlines, 3-D effects, unnecessary colour, and excessive labelling
- Tufte analysed **4,000 charts** from science, business, and journalism publications to identify patterns of effective and deceptive design
- His contemporaries focused on *how to draw* charts; Tufte focused on *what to show* and *what to remove*

Technical Contribution 1 — Data-Ink Ratio & Chartjunk

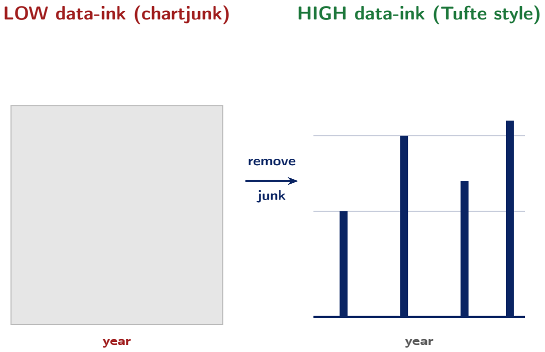
The Data-Ink Ratio

$$\text{Data-Ink Ratio} = \frac{\text{ink used to display data}}{\text{total ink used in the graphic}}$$

Goal: maximise data-ink ratio. Every non-data mark must justify its existence.

Chartjunk — Tufte's term for ink that does not convey data:

- **Vibrations**: optical illusions from cross-hatching and dense patterns
- **Grids**: heavy gridlines compete with data; use light or no gridlines
- **Ducks**: graphic elements that represent themselves rather than data (e.g., a chart drawn as a 3-D bar on a fake desktop)



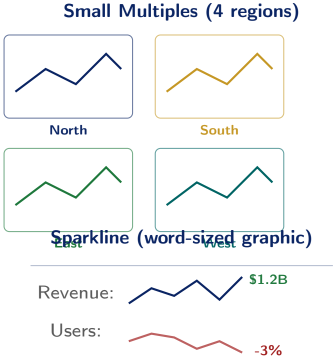
Technical Contribution 2 — Lie Factor, Small Multiples & Sparklines

The Lie Factor

$$\text{Lie Factor} = \frac{\text{size of effect shown in graphic}}{\text{size of effect in data}}$$

A lie factor of 1 is truthful. >1: graphic *overstates* the effect. <1: graphic *understates* it.

Classic example: a bar chart where the y-axis starts at 50 instead of 0. If the data effect is $+10\%$ but the bar appears to double in height, the lie factor is ≈ 10.



Small Multiples

The same chart structure repeated across many subsets of data.

Sparklines (Tufte 2006)

Sparklines are intense, simple, word-sized graphics embedded in text. Grafana and

Work 2 — Shneiderman (1996): The Eyes Have It

Shneiderman (1996): Publication Context & the Mantra

Full Citation

Shneiderman, B. (1996). **The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations.** In *Proceedings of the IEEE Symposium on Visual Languages*, pp. 336–343. Boulder, CO, USA.

Venue: IEEE Visual Languages — premier HCI/Viz venue
Cited by: >5,000 (Google Scholar, 2024)
Author: Ben Shneiderman, University of Maryland (inventor of the *treemap*, 1991)

The Visual Information Seeking Mantra

"Overview first,
zoom and filter

Seven Data Types

Type	**Example**
1-D (linear)	Time series, ranked lists
2-D (planar)	Geographic maps, scatter
3-D (volumetric)	MRI scans, CT volumes
Temporal	Timelines, Gantt charts
Multi-dimensional	Census microdata (10+ vars)
Tree	File system, org chart
Network	Social graph, dependency DAG

Seven Tasks (the same for every type)

Shneiderman's Treemap and Dynamic Queries

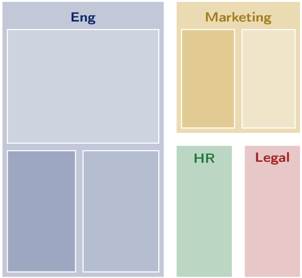
Treemap: Visualising Hierarchies as Area

Ben Shneiderman invented the **treemap** in 1991 (prior to the 1996 paper) to visualise a file system:

- Each file/directory is a rectangle
- **Area** $\propto$ file size
- **Colour** $\propto$ a second variable (modification date, file type)
- Hierarchy encoded by nesting (parent rectangle contains child rectangles)
- Enables inspection of **100,000+ nodes** at once — impossible with a tree diagram

Big Data applications: disk usage analysis (`WinDirStat`, `ncdu`), stock market heatmaps (Bloomberg, Finviz), NYT budget treemaps, Tableau built-in treemap chart type.

Treemap: disk usage by department



area = storage used; colour = growth rate

Dynamic Queries

Shneiderman proposed **dynamic queries**: sliders and buttons that update the visualisation in

Impact Today — Tufte (1983) & Shneiderman (1996)

How These Works Shape Modern Tools

Concept	**Modern instantiation**
Data-ink ratio	Tableau's "Remove formatting" one-click; Grafana's minimal theme
Chartjunk removal	D3's default: no axes, no gridlines (add only what you need)
Small multiples	Tableau's "Pages" shelf; Facet wrap in ggplot2; `facet_grid` in R
Sparklines	Grafana stat panel; Bloomberg terminal; GitHub contribution heatmap
Mantra	Tableau's drill-down;

Stanford CS246 / CMU 15-721 Perspective

- Stanford CS246 assigns Tufte as required reading for the **results communication** module — every data mining result is only as good as its visualisation
- CMU 15-721 uses Shneiderman's taxonomy to frame the **query interface design** for OLAP dashboards: a time-series drill-down is a 1-D $\rightarrow$ temporal transition in Shneiderman's taxonomy
- MIT 6.830 notes that Tufte's data-ink principle directly motivates **columnar storage**: retrieve only the columns ("data ink") needed; ignore the rest ("chartjunk")

Case Study — New York Times Graphics Desk (2010–2022)

The NYT Graphics Desk: Data Journalism at Scale

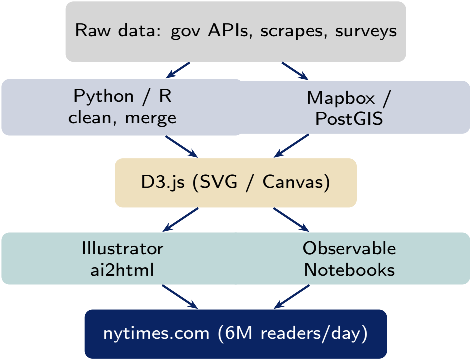
Who They Are and What They Build

The New York Times Graphics Desk is a team of ≈50 journalists, designers, and engineers who produce **interactive data visualisations** for nytimes.com — reaching **6 million readers per day**.

Notable members:

- **Mike Bostock** (2010–2015): invented **D3.js** while at NYT; used it to build the 2012 election maps
- **Amanda Cox**: Upshot editor; "You Draw It" participatory journalism pieces
- **Hannah Beachler**, **Josh Katz**: COVID-19 tracker team (2020–2022)

Their stack: **D3.js** for custom interactive graphics, **R** + **ggplot2** for exploratory analysis, **Python** +



Case: NYT COVID-19 Tracker (2020–2022)

The Design Challenge

From March 2020 the NYT Graphics Desk built and maintained the most-read COVID-19 data tracker in the world:

- **Data sources**: Johns Hopkins CSSE, CDC, 50 state health departments, county-level reporting — all with inconsistent schemas, missing values, and retroactive corrections
- **Daily pipeline**: Python scrapers → pandas cleaning → GitHub CSV repository → D3 charts auto-updated every hour
- **Charts produced**: 7-day rolling average line charts (Tufte small multiples for all 50 states), choropleth maps (county-level case rates), vaccination progress bar charts
- **Reach**: tracker page loaded **800 million times**

Tufte + Shneiderman Principles Applied

- **Data-ink ratio**: no gridlines on case charts; only a light horizontal reference at the prior peak
- **Small multiples**: 50-state chart grid lets readers compare all states simultaneously without animation
- **Overview first**: national total at the top; state breakdown below; county detail on click
- **Zoom and filter**: state selector dropdown (Shneiderman dynamic query); date range slider
- **Details-on-demand**: hover tooltip shows exact date, count, and 7-day change

Lessons Learned & Discussion Questions

Key Lessons from the NYT Case

1. **Data quality is a visualisation problem**: the NYT's biggest engineering effort was *cleaning* data, not drawing charts; retroactive state corrections forced D3 charts to support negative bars (days where cases were revised down)
2. **Accessibility is non-negotiable at 6M readers/day**: every chart had a data table alternative for screen readers; colour scales were tested for colour-blindness (deuteranopia, protanopia)
3. **Small multiples beat animation**: the 50-state grid (Tufte small multiples) was more actionable than an animated map of spreading cases
4. **Rolling averages are editorial decisions**: choosing 7-day vs. 14-day average changes *what*

Discussion Questions (CLO3)

1. A Grafana dashboard shows daily API error counts for a Big Data pipeline. Identify **three** Tufte violations in a dashboard that uses: a 3-D bar chart with a y-axis starting at 500, a heavy grid, and gradient fill colours. Redesign the chart applying data-ink principles. *[CLO3 — Apply]*
2. The NYT COVID tracker uses a **7-day rolling average** instead of raw daily counts. **(a)** What reporting artefact does this remove? **(b)** What real signal might a 7-day window mask that a 3-day window would reveal? Relate your answer to Shneiderman's "zoom and filter."

Live Exercise

Live Exercise — D3.js Bar Chart with Tufte Styling

Task (25 minutes, pairs)

Build a minimal D3.js bar chart that follows **Tufte's data-ink principles**:

1. No gridlines (or only a single light reference line)
2. y-axis starts at 0
3. No fill gradient — use a flat colour
4. Tooltip on hover (Shneiderman: details-on-demand)
5. Add a **sparkline** for each bar showing the 7-day trend

Dataset: weekly Kafka message counts for 5 topics (generate 7×5 random values).

Extension: add a dropdown filter to select one topic and zoom into its 7-day line chart.

HTML / D3 Skeleton

```
<!DOCTYPE html>
<meta charset="utf-8">
<style>
  .bar { fill: steelblue; }
  .bar:hover { fill: #b5651d; }
  .axis path, .axis line {
    stroke: #ccc; stroke-width: 0.5px; }
  .tooltip {
    position: absolute; background: #fff;
    border: 1px solid #ccc; padding: 6px;
    font-size: 11px; pointer-events: none; }
</style>
<body>
<script src=
  "https://d3js.org/d3.v7.min.js"></script>
<script>
const topics = ["orders","clicks",
               "errors","logins","search"];
const data = topics.map(t => ({
  topic: t,
  count: Math.floor(Math.random()*50000)
         + 10000
}));
```

References

Seminal Works

- Tufte, E. R. (1983; 2nd ed. 2001). *The Visual Display of Quantitative Information.* Graphics Press.
- Shneiderman, B. (1996). The eyes have it: A task by data type taxonomy for information visualizations. *Proc. IEEE Visual Languages*, 336–343.

Textbooks [SA], [TE], [LRU]

- **[SA]** Acharya, S., & Chellappan, S. (2015). *Big Data and Analytics* (Ch. 10 — Data Visualisation). Wiley.
- **[TE]** Erl, T., Khattak, W., & Buhler, P. (2016). *Big Data Fundamentals* (Ch. 11 — Visual Analytics). Prentice Hall.
- **[LRU]** Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). *Mining of Massive Datasets*, 3rd ed. (Ch. 1 — Data and visualisation). Cambridge UP.

Case Study Sources

- Bostock, M., Ogievetsky, V., & Heer, J. (2011). D^3: Data-driven documents. *IEEE Trans.*

Closing — Week 9, Part 2

Key Takeaways

- Tufte's **data-ink ratio** is a falsifiable metric: given two charts showing the same data, the one with the higher data-ink ratio communicates more efficiently. Apply it to every dashboard you build.
- Shneiderman's **mantra** (overview → zoom/filter → details) is a universal interaction design pattern — Tableau, Kibana, and D3 implement it differently but all adhere to it
- The NYT COVID tracker proved that **data quality is the invisible half of data visualisation**: the hardest engineering work was schema normalisation across 50 state health departments, not SVG rendering
- **D3.js** (created at NYT) is the direct application

Quote of the Week

"Above all else, show the data."

— Edward Tufte (1983)

"Overview first,
zoom and filter,
then details-on-demand."

— Ben Shneiderman (1996)

Part 1 Preview — Week 10

Social Media, Time Series & Health Data Analytics